Xiwen (Christina) Wei

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Education

PhD in Electrical and Computer Engineering *University of Texas at Austin* **Austin, TX** *Aug 2023-May 2027 (expect)*
GPA: 3.8/4.00 (GAIN 2025 Silver Industry Award, Cockrell School of Engineering PhD Fellowship)

BSE in Electrical Engineering *University of Michigan, Ann Arbor* **Ann Arbor, MI** *Aug 2021-May 2023*
GPA: 3.89/4.00 (Summa Cum Laude, Dean's List, James B. Angell Scholar)

BSE in Electrical and Computer Engineering *Shanghai Jiao Tong University* **Shanghai, China** *Sep 2019-Aug 2023*
Outstanding Graduate of Shanghai Jiao Tong University

Professional Experience

Graduate Research Assistant *System Level Design Group* **Austin, TX** *08/2023 - present*

- Developed continual-learning methods for **unified multimodal LLMs**, identifying and mitigating intra- and inter-modal forgetting across sequential tasks; published as first author at NeurIPS 2025 [1].
- Created Online-LoRA, a task-free online adaptation method for foundation vision models using **parameter-efficient low-rank updates**; published as first author at WACV 2025 [2].

Applied Scientist Intern *Zillow Group, Inc.* **Remote** *05/2026 - 08/2026*

- Developed an LLM-based user simulator for training and evaluating multi-turn conversational agents across 100 customer-service tasks spanning 10 domains.
- Designed a Turing-feedback-driven prompt-optimization pipeline, increasing win rate by **20.7–26.9** percentage points across three held-out LLM judges and **11.1** points under blinded human evaluation.
- Built end-to-end dialogue collection, annotation, and statistical evaluation tools, supporting 300 human–AI conversations and **900+** blinded human judgments.

PhD Research Associate *Advanced Micro Devices, Inc. (AMD)* **Austin, TX** *05/2025 - 12/2025*

- Led an end-to-end research project on **parameter-efficient post-training** of a **37B** unified multimodal model for interleaved text–image generation.
- Built distributed training and evaluation pipelines with **PyTorch, PEFT, ROCm, and Slurm**, running large-scale experiments across 8 AMD MI300X GPUs.
- Developed a Pareto-based gradient-balancing method to mitigate modality imbalance, improving image-generation quality by up to **19.3%** across multimodal benchmarks [4].

Research Fellow *University of Michigan Transportation Research Institute* **Ann Arbor, MI** *01/2022 - 04/2023*

- Developed statistical and neural models of 3D human anatomy from medical imaging data using **Python/MATLAB, PCA, and geometric analysis** to support personalized vehicle-safety research.

Selected Publications · [Full list]

[1] **Xiwen Wei**, Mustafa Munir, Radu Marculescu. *Mitigating Intra- and Inter-modal Forgetting in Continual Learning of Unified Multimodal Models*. The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS), 2025.

[2] **Xiwen Wei**, Guihong Li, Radu Marculescu. *Online-LoRA: Task-free Online Continual Learning via Low Rank Adaptation*. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025.

[3] Yuedong Yang, **Xiwen Wei**, Mustafa Munir, Radu Marculescu. *Fuel Gauge: Estimating the Length of Chain-of-Thought a priori for Large Multi-modality Models*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

[4] **Xiwen Wei**, Mark Nutter, Madhusudhanan Srinivasan, Radu Marculescu. *Pareto LoRA: Mitigating Modality Imbalance in Unified Multimodal Models via Pareto-Optimal Gradient Integration*. Preprint, 2026.

[5] **Xiwen Wei**, Guihong Li, Radu Marculescu. *Fairness Implications of Machine Unlearning: Bias Risks in Removing NSFW Content from Text-to-Image Models*, NeurIPS 2024 Workshop on Regulatable ML.

[6] Tianrui Hu, Dimitrios Liakopoulos, **Xiwen Wei**, Radu Marculescu, Neeraja J Yadwadkar. *Simulating rumor spreading in social networks using llm agents*, WMAC 2025: AAAI 2025 Workshop on Advancing LLM-Based Multi-Agent Collaboration.

Skills

- **Research**: Multimodal LLMs, Vision-language Models, LLM agents, post-training, continual learning, parameter-efficient fine-tuning, Generative AI, human evaluation
- **ML Systems**: PyTorch, Hugging Face Transformers, PEFT, distributed training, ROCm, vLLM, Docker, Slurm, Linux
- **Programming**: Python, C++, SQL, Bash, MATLAB
- **Data & Evaluation:** NumPy, Pandas, scikit-learn, statistical testing, human-subject evaluation